

# DArts

Diversity in the MoMA Collection

COM-480 Data Visualization - Milestone 3 Process Book

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Live: <https://com-480-data-visualization.github.io/DArts/>

Repository: <https://github.com/com-480-data-visualization/DArts>



*Final DArts website: scroll-driven narrative from collection shape to geography, gender, medium, and artist match.*

## 1. Motivation and question

MoMA is often imagined through canonical paintings and sculptures. The recorded collection tells a wider story: works on paper dominate the archive, artist nationality is geographically concentrated, gender representation changes slowly, and representation varies sharply by medium. Our central question is: **who is actually present in the recorded MoMA collection, and how does that presence change by medium, place, gender, and time?**

The target audience is broad: general visitors, students, and researchers who may not know art history. DArts is therefore a martini-glass narrative: authored scenes establish the main findings, the medium explorer opens into reader-driven filtering, and the quiz closes with a personal artist match.

## 2. Data

The data source is the public Museum of Modern Art Collection repository. We use artwork and artist CSV files, then aggregate them before runtime. The final site uses **144,149 cleaned artworks** and **11,879 cleaned artists**. A permissive parser extracts the first reliable year between 1700 and 2030 while keeping undated records rather than silently dropping them. Countries are mapped through a curated nationality-to-ISO3 lookup and collapsed to eight UN M49-inspired region buckets.

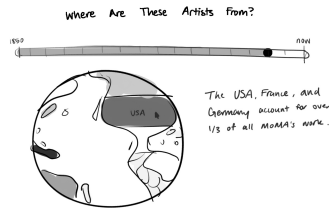
| Item                         | Count   | Design consequence                                      |
|------------------------------|---------|---------------------------------------------------------|
| Cleaned artworks             | 144,149 | Treemap percentages use the full cleaned corpus.        |
| Dated cleaned artworks       | 141,884 | Time filters exclude only records that cannot be dated. |
| Cleaned artists              | 11,879  | Artist match and country panels remain lightweight.     |
| Mapped artist credits        | 156,360 | Globe includes an explicit unmapped-nationality note.   |
| Unmapped nationality credits | 3,675   | Missing metadata is surfaced, not hidden.               |

### 3. Design process

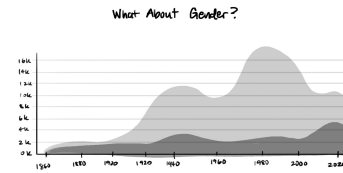
The initial sketches explored art-object metaphors: framed works for collection shape, an orthographic globe for origin, layered area/line charts for gender, small multiples for medium, and a hand-drawn quiz card for the personal ending. During implementation we kept the art-world tone but replaced encodings that would be hard to defend in COM-480: pie/donut sketches became 100% stacked bars, decorative artwork placeholders became metadata cards and MoMA links, and freeform annotations became leader-line callouts tied to data points.



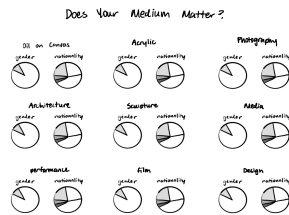
Scene 1 sketch



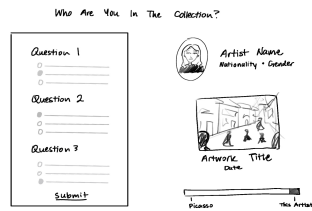
Scene 2 sketch



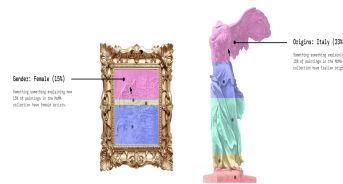
Scene 3 sketch



Scene 4 sketch



Scene 5 sketch



Annotation study

Final deltas: Scene 1 became a treemap for part-to-whole scale; Scene 2 retained the globe but added a legend and linked selection; Scene 3 switched to a fixed-scale line chart plus small multiples; Scene 4 replaced pies with stacked bars; Scene 5 became a deterministic underrepresented-artist match with a Picasso anchor bar.

## 4. Visualization choices

### Scene 1: The Collection Takes Shape

A treemap uses area for collection-area size. Area is less precise than position, but it is appropriate for a part-to-whole overview with few large categories. A single accent color highlights the active claim while the rest remains neutral, using Gestalt pop-out and Tufte data-ink restraint.

### Scene 2: Where Are These Artists From?

The map uses an orthographic projection, not Mercator. Country fill uses a sequential luminance ramp for count intensity. The United States, France, and Germany receive a hue exception as a deliberate pop-out for the top-three claim. Country click implements brushing and linking.

### Scene 3: What About Gender?

Female-credited share is encoded by vertical position over time, the strongest channel for quantitative comparison. The y-axis is fixed at 0-60% and labelled; the 50% parity line makes the meaningful reference point visible. Department small multiples keep shared scales.

### Scene 4: Does Your Medium Matter?

The explorer uses small multiples and 100% horizontal stacked bars for gender and region. We avoided pie charts because length on a common baseline is easier to compare than angles and small wedge areas.

### Scene 5: Who Are You In The Collection?

The quiz filters by decade, region, and medium, then selects an underrepresented artist deterministically. The anchor bar compares that artist to Picasso, using length on a shared scale for the emotional payload.

## 5. Interaction and narrative

The interaction model follows Shneiderman's mantra: overview first, zoom/filter, details on demand. Scenes 1-3 are scroll-driven and author-led; Scene 4 opens into filters and linked exploration; Scene 5 returns to a short personal story. D3 is used for scales, geo projection, treemap layout, and transitions, while Svelte renders marks declaratively. Motion is limited to state changes that support object constancy: treemap growth, line draw-on, globe rotation, and slider updates. The site respects prefers-reduced-motion.

## 6. Challenges and decisions

- ? Raw data reconciliation: the first preprocessing pass undercounted artworks because it required clean dates and several non-null fields. We kept rows with missing dates, parsed years permissively, and reconciled every aggregate against summary.json.
- ? Artwork images versus copyright: MoMA metadata is public, but artwork images are not automatically safe to embed. We link to MoMA records and avoid placing copyrighted images on the site.
- ? Globe legibility versus map accuracy: a choropleth globe is visually strong but interaction-heavy. We kept orthographic projection, added a legend and unmapped-note, and made the map a linked filter entry point.
- ? Pictorial sketches versus defensible encodings: early sketches used pie-like metaphors and art-object silhouettes. We retained the art tone through typography and motion but converted comparisons to treemaps, line charts, and stacked bars.
- ? Performance on 144k rows: the site never computes from raw CSV at runtime. Python pre-aggregates the data into JSON, and the larger artist index is loaded only when needed.
- ? Country fame ranking: the dataset has no true fame metric. We created a transparent collection-prominence proxy based on work count, temporal span, and department/medium breadth, while labelling it as a proxy.

## 7. Peer assessment

This is an editable draft split for the final submission.

| Member         | Share | Main contributions                                                                                          |
|----------------|-------|-------------------------------------------------------------------------------------------------------------|
| Oussama Ghali  | 34%   | Front-end integration, visual polish, globe interaction, motion design, deployment debugging, final QA.     |
| Nour Guermazi  | 33%   | Data cleaning support, EDA interpretation, gender/medium analysis, process-book drafting, accessibility r   |
| Isabella Linde | 33%   | Narrative structure, sketches and visual concept, quiz wording, design critique, screencast script support. |

## 8. References

Bertin, J. Semiology of Graphics. Brewer, C. A. ColorBrewer color schemes. Heer, J. and Robertson, G. Animated transitions in statistical data graphics. Heer, J. and Segel, E. Narrative visualization: telling stories with data. Munzner, T. Visualization Analysis and Design. Shneiderman, B. The eyes have it. Tufte, E. R. The Visual Display of Quantitative Information. Ware, C. Information Visualization: Perception for Design.